

Orienting response is modulated by the human-likeness and realism of the virtual proposer. Exploratory study with physiological measurement.

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ABSTRACT

This poster presents an exploratory analysis investigating the impact of virtual character's realism and human-likeness on participants' Orienting Response (OR) indexed by heart rate (HR) and skin conductance response (SCR). Fifty-nine participants, wearing HMD, watched the video recordings of the virtual characters and humans differing in terms of behavioral realism (movement and gaze), while their physiological responses were measured. Findings highlight a significant influence of behavioral realism on both indices of the OR (deeper HR deceleration and stronger SCR), which in case of SCR is further modulated by human-likeness of the virtual proposer.

Index Terms: Virtual Reality (VR)—psychology—orienting response—virtual character;

1 INTRODUCTION

Realism is considered a fundamental component of the virtual experience, impacting factors like presence, co-presence, immersion, and physiological responses [5]. It refers to the degree to which a virtual character is perceived as "real" [4] including aspects such as how it looks (photographic realism), behaves (behavioral realism), or interacts with the user (interactional realism) [7]. Drawing from the work of Syrjämäki et al. [8] and our recent study [5], we aimed to examine the impact of virtual character's behavioral realism on OR, measured using heart rate (HR) and skin conductance response (SCR). Moreover, we tested whether this manipulation would result in different effects depending on human-likeness [1], defined as the extent to which a virtual character simulates or bears resemblance to a human. This follows the findings of Syrjämäki et al. [8] and Kiilavuori et al. [2], which revealed distinct magnitudes of eye contact effects on the OR, depending on the human-likeness of the confederate (human vs. humanoid robot or virtual character).

2 PARTICIPANTS

A total of 59 participants (14 males) with an average age of 22.08 (SD = 3.85) took part in the study. They all had normal or corrected

to normal vision, no neurological diseases, no psychiatric disorders, and no susceptibility to motion sickness. Prior to the study, participants provided informed consent. Subjects were financially reimbursed for their involvement.

3 PROCEDURE

This exploratory study was part of a larger experiment where participants were led to believe they were playing the Ultimatum game against different types of proposers: computer, virtual characters with low or high behavioral realism and humans with low or high behavioral realism. Participants responded to different split propositions (50/50, 60/40, 70/30, 80/20, 90/10) of a 100 PLN.

Upon arrival at the laboratory, participants received study information and provided informed consent. Afterwards, they adjusted the VR head-mounted display and had HR and SCR activity sensors attached. After setup, participants entered the VR environment, familiarized themselves with instructions, and completed a training phase (5 trials). The main task involved playing the Ultimatum game with 5 proposers, responding to 5 propositions in 120 rounds (40 with the computer, 40 with virtual characters of varying behavioral realism, and 40 with humans of varying behavioral realism).

In this exploratory study, we are particularly interested in the proposer's presentation. We focus on this aspect, while the full trial structure is available in supplementary files on OSF. Each proposer's presentations (displayed for 6s) was preceded by a centrally located fixation point (displayed for 6-9s randomly). As, focus of the study was on investigating the impact of human vs. virtual character and their behavioral realism on OR, we excluded the computer as a proposer from analyses. Each participant's calculations were based on up to 80 trials (with an additional 0.5 percent of trials excluded due to missing data).

4 STIMULI

The stimuli covered two factors: human-likeness and behavioral realism. Human-likeness manipulation consisted of a comparison between human (a video of a real human) vs. virtual character (a recording of a virtual character). Manipulation of behavioral realism was built upon the eye contact and movement of the proposer (low - proposer not moving and not looking at participants; high - proposer moving and looking at participants). This amounted to four variants of proposers (human with low behavioral realism, human with high behavioral realism, virtual character with low behavioral realism and virtual character with high behavioral realism) which is available in supplementary files on OSF.

5 DATA PREPROCESSING

Please find a link to this part here:

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https://osf.io/cu57q/?view_only=deb30261d9d849cbb76e930da3c43eec

6 ANALYSES

Using SPSS ver. 24 (IBM, New York), we employed Generalized Estimating Equations (GEE) to explore the link between human-likeness, behavioral realism and physiological measures. Four separate analyses were conducted with different dependent variables: HR (0–2s, 2–4s, 4–6s) and EDA (SCR in the 1–4s time window) and human-likeness and behavioral realism as factorial predictors. A linear model with an identity link function was utilized.

Participants' ID served as a subject variable, while stimulus presentation order was designated as a within-subject variable. We employed AR(1) as a working correlation matrix structure and used robust variance estimator. The models encompassed main effects of predictors and their 2-way interaction.

7 RESULTS

The GEE analysis showed that SCR was affected by the human-likeness of the proposer and behavioral realism as well as their interaction. The SCR was significantly higher when a proposer was human than virtual character and had high behavioral realism in comparison to low behavioral realism.

Pairwise comparisons of the human-likeness and behavioral realism interaction showed that the human proposer with high behavioral realism induced significantly higher SCR than both human and virtual character with low behavioral realism (for both of these comparisons $p = .027$). Other pairwise comparisons were insignificant.

The HR in 2–4s and HR in 4–6s time windows were modulated by the behavioral realism of the proposer with deeper HR deceleration in high behavioral realism condition. Neither of the predictors modulated the HR in 0–2s time-window (see Table 1 in supplementary materials for detailed results).

8 DISCUSSION

Building on prior research exploring the influence of virtual characters' behavior on the OR (e.g., eye gaze in Syrjämäki et al. [8], physical contact in Swidrak and Pochwatko [6], approach and gaze in Wieser et al. [9], eye gaze and movement in Sterna et al. [5]), our study aimed to test the effects of behavioral realism and human-likeness manipulation on OR indexed by SCR and HR. We observed that realistically behaving proposers elicited increased arousal associated with the stimulus onset indexed by SCR, compared to those with no realistic behavior. Moreover, we not only demonstrated that gaze and movement manipulation can increase arousal, but also showed that these effects are enhanced by human-likeness: the highest arousal was induced in the condition with realistically behaving human. This aligns with our recent findings [5] where virtual character's high fidelity of visual appearance combined with high behavioral realism evoked highest OR arousal as indexed by SCR. In terms of HR as an index of OR, within two of the examined time windows, high behavioral realism induced deeper deceleration compared to the low behavioral realism. Stronger HR deceleration, in context of Lacey Lacey [3] intake-rejection model, is associated with deeper sensory intake of the stimuli. Therefore, our outcomes suggest that the manipulation of gaze and movement increases engagement with virtual proposers. This differs from our previous findings [5]) and Syrjämäki et al [8] who observed either no significant or meaningful effects of gaze manipulation on heart rate responses. The outcomes of this research highlight the impact of behavioral realism and the appearance of the virtual proposers on arousal and cognitive elaboration components of the OR, but need to be further confirmed in future studies as these analyses were performed in context of playing an Ultimatum game which might have increased overall emotional responses to the proposer's presentations.

9 SUPPLEMENTARY FILES

Supplementary files can be accessed through OSF:

https://osf.io/cu57q/?view_only=deb30261d9d849cbb76e930da3c43eec

10 ACKNOWLEDGMENTS

This work was financed from the budget for science in the years 2019–2023, as a research project “The impact of virtually generated characters: human aspects of bots” (project number: DI2018 015848) under the Diamond Grant programme granted by the Ministry of Science and Higher Education of Poland. Work of M. Igras-Cybulska was supported by program Lider grant nr 0230/L-11/2019 by the National Center for Research and Development, Poland. The research has been supported by a Visibility and Mobility module and grants supporting publications in top journals and publishers under the Strategic Programme Excellence Initiative at the Jagiellonian University.

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